

Q20 Potential at X is 12 V

The p.d. between X and Y = p.d. across the $3\ \Omega$ resistor = $(1)(3) = 3\ \text{V}$

So the potential at Y is $12 - 3 = 9\ \text{V}$

Answer: D

Q21 By Fleming's left-hand rule, E is perpendicular to B